Chapter 4

Structural Properties of Linear Systems

After this learning unit you will know the definition of the three properties of linear systems in state space, whose investigation is essential for control engineering: asymptotic stability, controllability, and observability. You will also learn methods to verify these properties for a given system.

4.1 Asymptotic Stability

The objective of this learning unit is to understand the concept of asymptotic stability of a system as well as the relationship between stability and the eigenmotion of a system. At the end of the learning unit you should be able to examine a given linear system for stability using the presented methods.

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Asymptotic stability is a property of the solution of a dynamic system. It is closely linked to the results of the previous section, however it is not necessary to know the solution of a linear dynamic system in order to make statements about its asymptotic stability.

4.1.1 Meaning of the Eigenvectors

Eigenvectors

The Eigenvectors of the system matrix $\boldsymbol{A}$, which were already important for the transformation to diagonal form or to the Jordan normal form in Chapter 3, can be interpreted in a geometrically intuitive way. For this purpose, an autonomous system is considered

$$\dot{x} = \boldsymbol{A}\boldsymbol{x}\,, \quad x(0) = x_0 \tag{4.1}$$

and an arbitrary eigenvector $\boldsymbol{v}_i$ is considered according to its definition $\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}$. If it is now assumed that the initial state $\boldsymbol{x}_0$ of system (4.1) lies in the direction of $\boldsymbol{v}_i$

$$\boldsymbol{x}_0 = c_0 \boldsymbol{v}_i\,, \quad c_0 \in \mathbb{R}\,, \tag{4.2}$$

then the trajectory $\boldsymbol{x}(t)$ will move along the eigenvector $\boldsymbol{v}_i$:

$$\boldsymbol{x}(t) = c(t)\boldsymbol{v}_i \,. \tag{4.3}$$

This becomes apparent when one substitutes (4.3) into the system equation (4.1)

$$\dot{c}(t)\boldsymbol{v}_i = c(t)\boldsymbol{A}\boldsymbol{v}_i = c(t)\lambda_i\boldsymbol{v}_i \,. \tag{4.4}$$

With $\boldsymbol{v}_i \neq 0$ one obtains the differential equation

$$\dot{c}(t) = c(t)\lambda_i \quad \text{or} \quad c(t) = c_0 e^{\lambda_i t} \,. \tag{4.5}$$

It is seen that the original system (4.1) can thus be described along an eigenvector $\boldsymbol{v}_i$ by a differential equation for the amplification factor $c(t)$. The corresponding solution $\boldsymbol{x}(t) = c_0 e^{\lambda_i t}\boldsymbol{v}_i$ is also called the (i-th) *eigenmotion* (or *mode*) of the system (4.1). The eigenvalue λ_i indicates the temporal direction of motion along the eigenvector $\boldsymbol{v}_i$.

Eigenmotion

Example 4.1. *As an example, consider the following second order system*

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} = \begin{bmatrix} 2 & -1 \\ -1 & 0 \end{bmatrix} \boldsymbol{x} \,. \tag{4.6}$$

The eigenvalues of the dynamics matrix $\boldsymbol{A}$ *are calculated as*

$$\lambda_1 = 1 - \sqrt{2} \approx -0.414 \,, \qquad \lambda_2 = 1 + \sqrt{2} \approx 2.414$$

with the corresponding eigenvectors

$$\boldsymbol{v}_1 = \begin{bmatrix} -1+\sqrt{2} \\ 1 \end{bmatrix} \approx \begin{bmatrix} 0.414 \\ 1 \end{bmatrix} , \quad \boldsymbol{v}_2 = \begin{bmatrix} -1-\sqrt{2} \\ 1 \end{bmatrix} \approx \begin{bmatrix} -2.414 \\ 1 \end{bmatrix} . \tag{4.7}$$

The family of trajectories of system (4.6) *is shown together with the eigenvectors* $\boldsymbol{v}_1$ *and* $\boldsymbol{v}_2$ *in Figure 4.1. It is clearly seen that trajectories that move in the direction of* $\boldsymbol{v}_1$ *converge to the origin* ($\lambda_1 < 0$) *and in the direction of* $\boldsymbol{v}_2$ *diverge from the origin* ($\lambda_2 > 0$).

4.1.2 Asymptotic Stability

With the previous investigations it is now possible to examine the stability behavior of a linear, time-invariant, autonomous system

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} \,, \quad \boldsymbol{x}(0) = \boldsymbol{x}_0 \,. \tag{4.8}$$

Asymptotic stability is defined as follows:

Definition 4.1 (Asymptotic Stability). *The linear system* (4.8) *is called asymptotically stable if for all* $\boldsymbol{x}(0) = \boldsymbol{x}_0 \neq 0$ *the limit* $\lim_{t\to\infty} ||\boldsymbol{x}(t)|| = 0$ *holds.*

Definition asymptotic stability

The solution of (4.8) involves the transition matrix $\Phi(t)$, see equation (3.43). In the investigation of asymptotic stability

$$||\boldsymbol{x}(t)|| = ||\Phi(t)\boldsymbol{x}_0|| \leq ||\Phi(t)|| \cdot ||\boldsymbol{x}_0|| \tag{4.9}$$

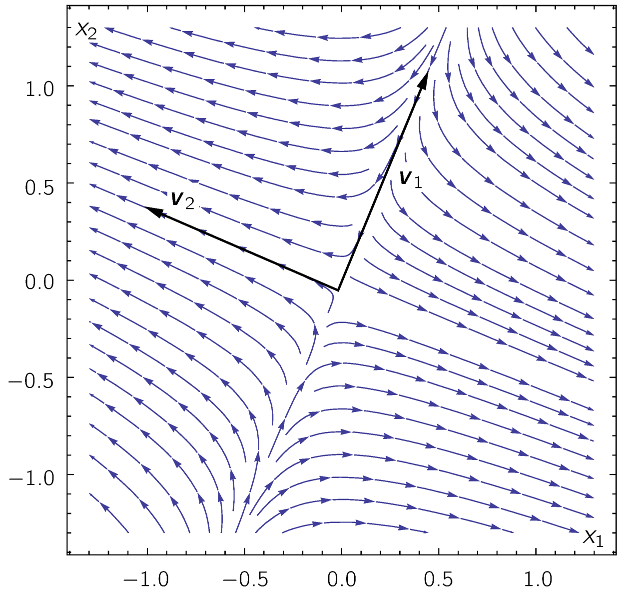

Figure 4.1: Trajectories and eigenvectors of system (4.6) (Example 4.1).

the induced matrix norm $||\Phi(t)||$ of the transition matrix plays an important role.
To simplify matters, it is initially assumed that the system (4.1) has distinct eigenvalues λ_i, so that the solution $\boldsymbol{x}(t) = \Phi(t)\boldsymbol{x}_0$ is determined via the diagonal form

$$\boldsymbol{x}(t) = \Phi(t)\boldsymbol{x}_0 = \boldsymbol{V}\tilde{\Phi}(t)\boldsymbol{V}^{-1}\boldsymbol{x}_0 = \boldsymbol{V}\begin{bmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{\lambda_n t} \end{bmatrix}\boldsymbol{V}^{-1}\boldsymbol{x}_0$$

Obviously, it holds that

$$\lim_{t\to\infty} \left|e^{\lambda_i t}\right| = \begin{cases} 0 & \text{for } \mathrm{Re}(\lambda_i) < 0 \\ 1 & \text{for } \mathrm{Re}(\lambda_i) = 0 \\ \infty & \text{for } \mathrm{Re}(\lambda_i) > 0 \end{cases}$$

Thus, $||\boldsymbol{x}(t)||$ tends to 0 if and only if $\mathrm{Re}(\lambda_i) < 0$ holds. For repeated eigenvalues it follows from (3.71) that for the of the transition matrix $\tilde{\Phi}(t)$

$$\lim_{t\to\infty} \left|t^k e^{\lambda_i t}\right| = \begin{cases} 0 & \text{for } \mathrm{Re}(\lambda_i) < 0 \\ 1 & \text{for } \mathrm{Re}(\lambda_i) = 0 \quad (\text{if } \lambda_i \text{ is simple}) \\ \infty & \text{for } \mathrm{Re}(\lambda_i) = 0 \quad (\text{if } \lambda_i \text{ is repeated}) \\ \infty & \text{for } \mathrm{Re}(\lambda_i) > 0 \end{cases}$$

must hold. Thus, the condition $\mathrm{Re}(\lambda_i) < 0$ for the asymptotic stability of $||\boldsymbol{x}(t)|| = ||\boldsymbol{V}\tilde{\Phi}(t)\boldsymbol{V}^{-1}|| \cdot ||\boldsymbol{x}_0||$ is also valid for repeated eigenvalues.
These considerations can also be transferred to conjugate complex eigenvalues, see (3.68). In general, with regard to the Jordan normal form in (3.23)–(3.25) the following theorem holds:

Theorem 4.1 (Solution Behavior). *Each component of the solution $\boldsymbol{x}(t) = [x_1(t), \ldots, x_n(t)]^\mathsf{T}$ of the system (4.8) is a linear combination of functions of the form*

$$t^k e^{\lambda t}\,, \quad t^k e^{\alpha t}\cos(\beta t)\,, \quad t^k e^{\alpha t}\sin(\beta t) \tag{4.10}$$

depending on the real or conjugate complex eigenvalues $\lambda = \alpha \pm j\beta$ and with constants $0 \leq k \leq n-1$.

As a direct consequence of Theorem 4.1, the conditions for asymptotic stability can be stated as follows:

Conditions for asymptotic stability

Theorem 4.2 (Asymptotic Stability). *The equilibrium $\boldsymbol{x}_R = 0$ of the linear system (4.8) is asymptotically stable if and only if the matrix $\boldsymbol{A}$ is a so-called Hurwitz matrix, i.e.*, if all eigenvalues have a negative real part, *i.e.* $\mathrm{Re}(\lambda_i) < 0$.

In Figure 4.2 some examples of eigenvalues are shown in order to illustrate the general solution behavior:

a) The time responses decay monotonically, the system is asymptotically stable. The slowest process (eigenvalue λ_1) dominates the time response.

b) The system is unstable; with an initial disturbance $|x(t)|$ diverges to infinity without oscillations.

c) The system is asymptotically stable, the time response has oscillatory components.

d) The system is unstable, with rising oscillatory time responses.

Question 4.1. *How can one examine a system for stability?*

Exercise 4.1. *Determine whether the equilibrium $\boldsymbol{x}_R = 0$ of the following systems*

$$\dot{\boldsymbol{x}} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & -3 & -2 \end{bmatrix} \boldsymbol{x}\,, \qquad \dot{\boldsymbol{x}} = \begin{bmatrix} -1 & 0 \\ -2 & 0 \end{bmatrix} \boldsymbol{x}$$

is asymptotically stable.

Exercise 4.2. *Calculate the eigenvalues for the linearized model (2.47) of the cart-pendulum system in Exercise 2.6 and state what follows therefrom regarding the stability of the system in the upper equilibrium.*

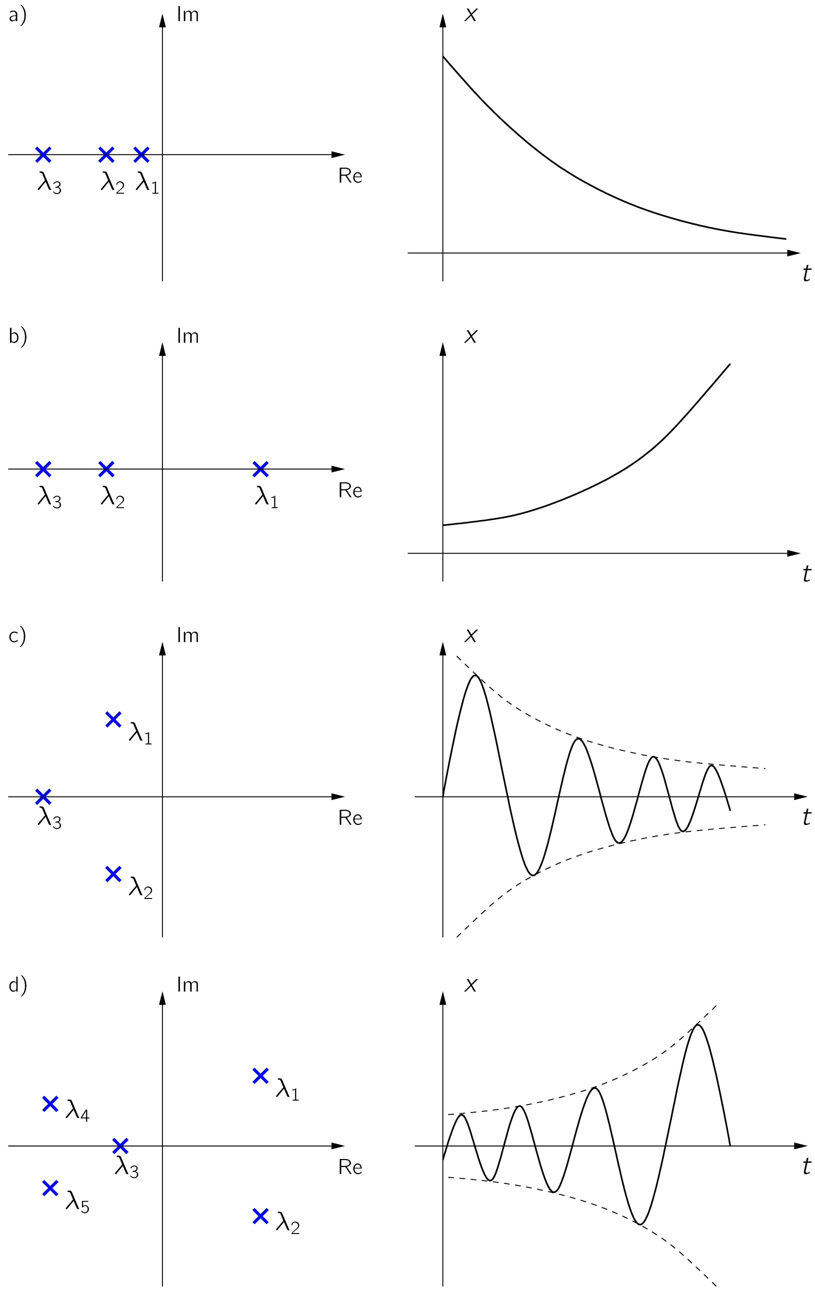

Figure 4.2: Solution behavior for different eigenvalues.

4.2 Controllability

In this learning unit you will learn the definition of controllability according to Kalman, the formula for the controllability matrix, and the control canonical form. At the end of the learning unit you should be able to examine a given system for controllability and transform it into control canonical form.

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The concept of controllability is explained here for a linear, time-invariant multi-input multi-output system

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}\,, \quad \boldsymbol{x}(0) = \boldsymbol{x}_0 \tag{4.11}$$

with the state $\boldsymbol{x} \in \mathbb{R}^n$, the input $\boldsymbol{u} \in \mathbb{R}^m$ and the matrices $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{B} \in \mathbb{R}^{n\times m}$.

Definition Controllability

Definition 4.2 (Controllability). *The linear system* (4.11) *is called* (completely) controllable *if it can be transferred in finite time* $T < \infty$ *from any initial state* $\boldsymbol{x}_0$ *by a piecewise continuous control signal* $\boldsymbol{u}(t)$*,* $t \in [0, T]$ *to any desired final state* $\boldsymbol{x}(T) = 0$.

4.2.1 Controllability According to Kalman

To examine the property of controllability of the linear system (4.11), one first considers the general solution according to (3.56)

$$\boldsymbol{x}(t) = \Phi(t)\boldsymbol{x}_0 + \int_0^t \Phi(t-\tau)\boldsymbol{B}\boldsymbol{u}(\tau)\,\mathrm{d}\tau \tag{4.12}$$

with the transition matrix $\Phi(t)$ from (3.45). Without loss of generality, one considers the initial state $\boldsymbol{x}_0 = 0$ and sets $t = T$:

$$\begin{aligned}\boldsymbol{x}(T) &= \int_0^T \Phi(T-\tau)\boldsymbol{B}\boldsymbol{u}(\tau)\,\mathrm{d}\tau = \int_0^T \sum_{k=0}^{\infty} \boldsymbol{A}^k \frac{(T-\tau)^k}{k!}\boldsymbol{B}\boldsymbol{u}(\tau)\,\mathrm{d}\tau \\ &= \sum_{k=0}^{\infty} \boldsymbol{A}^k\boldsymbol{B}\hat{\boldsymbol{u}}_k \quad \text{with} \quad \hat{\boldsymbol{u}}_k = \int_0^T \frac{(T-\tau)^k}{k!}\boldsymbol{u}(\tau)\,\mathrm{d}\tau\,.\end{aligned} \tag{4.13}$$

For an arbitrary control trajectory $\boldsymbol{u}(t)$, $t \in [0, T]$ with fixed final time T, the quantities $\hat{\boldsymbol{u}}_k$ are constant m-dimensional vectors. From (4.13) it is evident that $\boldsymbol{x}(T)$ can be represented as a linear combination of the column vectors of $\boldsymbol{B}, \boldsymbol{A}\boldsymbol{B}, \boldsymbol{A}^2\boldsymbol{B}, \ldots$.

necessary condition for controllability

A necessary condition for controllability is therefore that the column vectors of $\boldsymbol{B}, \boldsymbol{A}\boldsymbol{B}, \boldsymbol{A}^2\boldsymbol{B}, \ldots$ span $\mathbb{R}^n$. In other words, the matrix

$$\left[\boldsymbol{B}\,,\ \boldsymbol{A}\boldsymbol{B}\,,\ \boldsymbol{A}^2\boldsymbol{B}\,,\ \ldots\,,\ \boldsymbol{A}^{n-1}\boldsymbol{B}\,,\ \boldsymbol{A}^n\boldsymbol{B}\,,\ \boldsymbol{A}^{n+1}\boldsymbol{B}\,,\ \ldots\right] \tag{4.14}$$

must have rank n. To continue this intermediate result, the *Cayley-Hamilton Theorem* (Theorem 3.1) can be used:
Since the matrices $\boldsymbol{A}^n\boldsymbol{B}$, $\boldsymbol{A}^{n+1}\boldsymbol{B}, \ldots$ according to Theorem 3.1 obviously do not provide any further linearly independent entries in the matrix (4.14), it is sufficient in the rank verification of (4.14) to consider only the matrix entries from $\boldsymbol{B}$ up to $\boldsymbol{A}^{n-1}\boldsymbol{B}$. These considerations lead to the following theorem:

Theorem 4.3 (Kalman Controllability Criterion)**.** *The system* (4.11) *or, equivalently, the matrix pair* $(\boldsymbol{A}, \boldsymbol{B})$ *is* completely controllable *if and only if the* controllability matrix

$$\boldsymbol{Q}_S = \left[\boldsymbol{B}\,,\ \boldsymbol{A}\boldsymbol{B}\,,\ \ldots\,,\ \boldsymbol{A}^{n-1}\boldsymbol{B}\right] \ \text{or}\ \boldsymbol{Q}_S = \left[\boldsymbol{b}\,,\ \boldsymbol{A}\boldsymbol{b}\,,\ \ldots\,,\ \boldsymbol{A}^{n-1}\boldsymbol{b}\right] \tag{4.15}$$

has rank n*.*

Kalman Controllability Criterion

To show that Theorem 4.3 is also *sufficient*, i.e. that the rank condition $\mathrm{Rang}(\boldsymbol{Q}_S) = n$ implies controllability, one considers the so-called *Gramian Matrix (controllability gramian)*

$$\boldsymbol{G} = \int_0^{\mathrm{T}} e^{\boldsymbol{A}\tau}\boldsymbol{B}\boldsymbol{B}^{\mathrm{T}}\left(e^{\boldsymbol{A}\tau}\right)^{\mathrm{T}} \mathrm{d}\tau \tag{4.16}$$

with $e^{\boldsymbol{A}t} = \Phi(t)$. Using similar arguments as above, one can show that $\boldsymbol{G}$ is nonsingular for all times $T > 0$ if $\mathrm{Rang}(\boldsymbol{Q}_S) = n$ is satisfied. Thus, for a desired time T, by substituting the control function

$$\boldsymbol{u}(t) = \boldsymbol{B}^{\mathrm{T}}\left(e^{\boldsymbol{A}(T-t)}\right)^{\mathrm{T}} \boldsymbol{G}^{-1}\left(\boldsymbol{x}(T) - e^{\boldsymbol{A}T}\boldsymbol{x}_0\right) \tag{4.17}$$

into equation (4.12)

$$\begin{aligned}
\boldsymbol{x}(T) &= \Phi(T)\boldsymbol{x}_0 + \int_0^{\mathrm{T}} \Phi(T-\tau)\boldsymbol{B}\boldsymbol{u}(\tau)\mathrm{d}\tau \\
&= \Phi(T)\boldsymbol{x}_0 + \int_0^{\mathrm{T}} \Phi(\tau)\boldsymbol{B}\boldsymbol{u}(T-\tau)\mathrm{d}\tau \\
&= \Phi(T)\boldsymbol{x}_0 + \underbrace{\int_0^{\mathrm{T}} e^{\boldsymbol{A}\tau}\boldsymbol{B}\boldsymbol{B}^{\mathrm{T}}\left(e^{\boldsymbol{A}\tau}\right)^{\mathrm{T}} \mathrm{d}\tau}_{=\boldsymbol{G}}\, \boldsymbol{G}^{-1}\left(\boldsymbol{x}(T) - e^{\boldsymbol{A}T}\boldsymbol{x}_0\right) \\
&= \boldsymbol{x}(T)
\end{aligned} \tag{4.18}$$

and thereby show that the system (4.11) is driven to the desired state $\boldsymbol{x}(T)$. If the system is controllable, then there exist infinitely many controls to steer the system (4.11) from $\boldsymbol{x}_0$ to $\boldsymbol{x}_T$. An interesting aspect of the specific control (4.17) is that it realizes the transition over the time interval $[0, T]$ with minimal energy and thus minimizes the following cost functional:

Minimum Energy Control

$$J(\boldsymbol{u}) = \int_0^{T} \boldsymbol{u}^{\mathrm{T}}(\tau)\boldsymbol{u}(\tau)\,\mathrm{d}\tau\,.$$

Example 4.2 (Cart-Pendulum System)**.** *Consider again the inverted pendulum depicted in Figure 2.3 with the nonsingular controllability matrix (4.19). The inverse controllability matrix is*

$$\boldsymbol{Q}_S = \begin{bmatrix} 0 & \frac{-6}{l(m+4M)} & 0 & \frac{-36g(m+M)}{l^2(m+4M)^2} \\ \frac{-6}{l(m+4M)} & 0 & \frac{-36g(m+M)}{l^2(m+4M)^2} & 0 \\ 0 & \frac{4}{m+4M} & 0 & \frac{18mg}{l(m+4M)^2} \\ \frac{4}{m+4M} & 0 & \frac{18mg}{l(m+4M)^2} & 0 \end{bmatrix} \tag{4.19}$$

with $\det(\boldsymbol{Q}_S) = \frac{1296\,g^2}{l^4(m+4M)^4} \neq 0$. *Since* $\boldsymbol{Q}_S$ *is nonsingular, the linearized pendulum model in the upper equilibrium is controllable.*

Question 4.2. *What does the Kalman controllability criterion state?*

Exercise 4.3. *Calculate the controllability matrix for the following system and state whether the system is controllable:*

$$\dot{x} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & -3 & -2 \end{bmatrix} \boldsymbol{x} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \boldsymbol{x}$$

For higher-dimensional problems the controllability analysis using $\boldsymbol{Q}_S$ will generally become increasingly poorly conditioned. As an alternative, one may verify the Gramian matrix (4.16):

Controllability via the Gramian Matrix

Theorem 4.4 (Controllability via the Gramian Matrix)**.** *The system* (4.11) *is* (completely) controllable *if and only if the* Gramian Matrix (controllability gramian)

$$\boldsymbol{G} = \int_0^T e^{\boldsymbol{A}\tau} \boldsymbol{B}\boldsymbol{B}^{\mathsf{T}} \left(e^{\boldsymbol{A}\tau}\right)^{\mathsf{T}} \mathrm{d}\tau \tag{4.20}$$

is nonsingular. If the dynamics matrix $\boldsymbol{A}$ *is a Hurwitz matrix, then* $\boldsymbol{G}_\infty = \lim_{T\to\infty} \boldsymbol{G}$ *can be computed from the following equation (a so-called Lyapunov equation):*

$$\boldsymbol{A}\boldsymbol{G}_\infty + \boldsymbol{G}_\infty \boldsymbol{A}^{\mathsf{T}} + \boldsymbol{B}\boldsymbol{B}^{\mathsf{T}} = \boldsymbol{0}\,. \tag{4.21}$$

If $\boldsymbol{A}$ is a *Hurwitz matrix* (i.e. all eigenvalues have negative real parts), then the existence of the integral (4.20) for the limit case $T \to \infty$ is guaranteed. In Matlab, the controllability matrix $\boldsymbol{Q}_S$ and the Gramian Matrix $\boldsymbol{G}_\infty$ can be calculated using the functions ctrb and gram.

Question 4.3. *Name two methods for examining the controllability of a system.*

4.2.2 Control Canonical Form (Single-Input Case)

In the following, the linear single-input system

$$\begin{aligned} \dot{\boldsymbol{x}} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}u\,, \quad \boldsymbol{x}(0) = \boldsymbol{x}_0 \\ y &= \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} + du \end{aligned} \tag{4.22}$$

with state $\boldsymbol{x} \in \mathbb{R}^n$, the scalar input $u \in \mathbb{R}$, and the scalar output $y \in \mathbb{R}$ is considered. This section will show how a linear system (4.22) in state space can be transformed into the control canonical form

$$
\dot{\tilde{x}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix}}_{\tilde{\boldsymbol{A}} = \boldsymbol{T}\boldsymbol{A}\boldsymbol{T}^{-1}} \tilde{x} + \underbrace{\begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}}_{\tilde{\boldsymbol{b}} = \boldsymbol{T}\boldsymbol{b}} u
\tag{4.23}
$$

$$
y = \begin{bmatrix} b_0 - a_0 b_n & \dots & b_{n-1} - a_{n-1} b_n \end{bmatrix} \tilde{x} + b_n u
$$

via a nonsingular state transformation

$$
\boldsymbol{x} = \boldsymbol{V}\tilde{\boldsymbol{x}} \quad \text{or} \quad \tilde{\boldsymbol{x}} = \boldsymbol{T}\boldsymbol{x} \quad \text{with} \quad \boldsymbol{T} = \boldsymbol{V}^{-1} \tag{4.24}
$$

into control canonical form. The controllability matrix $\boldsymbol{Q}_S$ plays an important role here.
From (4.23) it is evident that the first state $\tilde{x}_1$ of $\tilde{\boldsymbol{x}} = [\tilde{x}_1, \dots, \tilde{x}_n]^\mathsf{T}$ can be differentiated n times before the input u appears. In the following, $\tilde{x}_1$ is set by the equation

$$
\tilde{x}_1 = \boldsymbol{t}^\mathsf{T}\boldsymbol{x} \tag{4.25}
$$

where the vector $\boldsymbol{t}$ is to be determined such that the input u appears only in the n-th time derivative. Successive differentiation of $\tilde{x}_1$ with respect to time t and comparison with (4.23) yields

$$
\begin{aligned}
\dot{\tilde{x}}_1 &= \tilde{x}_2 = \boldsymbol{t}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{t}^\mathsf{T}\boldsymbol{b}\,u & \text{with} \qquad \boldsymbol{t}^\mathsf{T}\boldsymbol{b} &\overset{!}{=} 0 \\
\dot{\tilde{x}}_2 &= \tilde{x}_3 = \boldsymbol{t}^\mathsf{T}\boldsymbol{A}^2\boldsymbol{x} + \boldsymbol{t}^\mathsf{T}\boldsymbol{A}\boldsymbol{b}\,u & \boldsymbol{t}^\mathsf{T}\boldsymbol{A}\boldsymbol{b} &\overset{!}{=} 0 \\
&\vdots \\
\tilde{x}_1^{(n-1)} &= \tilde{x}_n = \boldsymbol{t}^\mathsf{T}\boldsymbol{A}^{n-1}\boldsymbol{x} + \boldsymbol{t}^\mathsf{T}\boldsymbol{A}^{n-2}\boldsymbol{b}\,u & \boldsymbol{t}^\mathsf{T}\boldsymbol{A}^{n-2}\boldsymbol{b} &\overset{!}{=} 0 \\
\tilde{x}_1^{(n)} &= \dot{\tilde{x}}_n = \boldsymbol{t}^\mathsf{T}\boldsymbol{A}^{n}\boldsymbol{x} + \boldsymbol{t}^\mathsf{T}\boldsymbol{A}^{n-1}\boldsymbol{b}\,u & \boldsymbol{t}^\mathsf{T}\boldsymbol{A}^{n-1}\boldsymbol{b} &\overset{!}{=} 1\,.
\end{aligned}
\tag{4.26}
$$

The constraints can be written in vector form as

$$
\boldsymbol{t}^\mathsf{T} \underbrace{[\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \dots, \boldsymbol{A}^{n-1}\boldsymbol{b}]}_{\boldsymbol{Q}_S} = [0, \dots, 0, 1]
$$

in dependence on the controllability matrix $\boldsymbol{Q}_S$. If $\boldsymbol{Q}_S$ is nonsingular, it follows that

$$
\boldsymbol{t}^\mathsf{T} = [0, \dots, 0, 1]\boldsymbol{Q}_S^{-1}\,. \tag{4.27}
$$

Evidently, the first state $\tilde{x}_1 = \boldsymbol{t}^\mathsf{T}\boldsymbol{x}$ of the control canonical form is determined by the last row of the inverse controllability matrix $\boldsymbol{Q}_S^{-1}$. Thus, the following theorem holds:

Transformation to Control Canonical Form

Theorem 4.5 (Transformation to Control Canonical Form)**.** *The linear system (4.22) can be transformed into the control canonical form*

(4.23) by the transformation (4.24) with the transformation matrix

$$\boldsymbol{T} = \begin{bmatrix} \boldsymbol{t}^{\mathsf{T}} \\ \boldsymbol{t}^{\mathsf{T}}\boldsymbol{A} \\ \vdots \\ \boldsymbol{t}^{\mathsf{T}}\boldsymbol{A}^{n-1} \end{bmatrix}, \quad \boldsymbol{t}^{\mathsf{T}} = [0, \dots, 0, 1]\boldsymbol{Q}_S^{-1}, \quad \boldsymbol{Q}_S = [\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \dots, \boldsymbol{A}^{n-1}\boldsymbol{b}] \tag{4.28}$$

provided that the controllability matrix $\boldsymbol{Q}_S$ is nonsingular. Here, $\boldsymbol{t}^{\mathsf{T}}$ represents the last row of the inverse controllability matrix.

An advantage of the control canonical form is that it can be easily determined from a transfer function $G(s)$. Furthermore, it is characteristic that the input u affects only the last state of $\tilde{\boldsymbol{x}}$, which is particularly advantageous for controller design.
The control canonical form can also be given for multi-input systems (4.11). This will be treated in the next chapter in the context of state-feedback controller design.

Example 4.3 (Cart-Pendulum System)**.** *Again, consider the inverted pendulum shown in Figure 2.3 with the nonsingular controllability matrix (4.19). The inverse controllability matrix is given by*

$$\boldsymbol{Q}_S^{-1} = \begin{bmatrix} 0 & \frac{ml}{2} & 0 & m+M \\ \frac{ml}{2} & 0 & m+M & 0 \\ 0 & -\frac{l^2(m+4M)}{9g} & 0 & -\frac{l(m+4M)}{6g} \\ -\frac{l^2(m+4M)}{9g} & 0 & -\frac{l(m+4M)}{6g} & 0 \end{bmatrix}. \tag{4.29}$$

With the last row

$$\boldsymbol{t}^{\mathsf{T}} = \begin{bmatrix} -\frac{l^2(m+4M)}{9g} & 0 & -\frac{l(m+4M)}{6g} & 0 \end{bmatrix} \tag{4.30}$$

and the dynamics matrix from (2.47) the transformation matrix (4.28)

$$\boldsymbol{T} = \begin{bmatrix} -\frac{l^2(m+4M)^2}{9g} & 0 & -\frac{l(m+4M)}{6g} & 0 \\ 0 & -\frac{l^2(m+4M)^2}{9g} & 0 & -\frac{l(m+4M)}{6g} \\ -\frac{l(m+4M)}{6} & 0 & 0 & 0 \\ 0 & -\frac{l(m+4M)}{6} & 0 & 0 \end{bmatrix} \tag{4.31}$$

and finally the control canonical form (4.23)

$$\dot{\tilde{\boldsymbol{x}}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{6g(m+M)}{l(m+4M)} & 0 \end{bmatrix} \tilde{\boldsymbol{x}} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u\,. \tag{4.32}$$

4.3 Observability

In most control engineering applications not all state variables can be measured. Observability of a system therefore deals with the question whether the system state can be reconstructed from the knowledge of the measured outputs $\boldsymbol{y}(t)$ and, if applicable, the inputs $\boldsymbol{u}(t)$.

In this learning unit you will learn the definition of observability according to Kalman, as well as the formula for the observability matrix and the observer canonical form. At the end of the learning unit you should be able to examine a given system for observability and transform it into observer canonical form.

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In further continuation, a linear, time-invariant multi-input multi-output system

$$\begin{aligned}\dot{\boldsymbol{x}} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}\,, \quad \boldsymbol{x}(0) = \boldsymbol{x}_0 \\ \boldsymbol{y} &= \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u}\end{aligned} \tag{4.33}$$

with the state $\boldsymbol{x} \in \mathbb{R}^n$, the input $\boldsymbol{u} \in \mathbb{R}^m$, the output $\boldsymbol{y} \in \mathbb{R}^p$, and the matrices $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\boldsymbol{B} \in \mathbb{R}^{n\times m}$, $\boldsymbol{C} \in \mathbb{R}^{p\times n}$ and $\boldsymbol{D} \in \mathbb{R}^{p\times m}$ is considered.

Definition 4.3 (Observability). *The system* (4.33) *is called (completely) observable if the initial state* $\boldsymbol{x}_0$ *can be determined from the knowledge of the input and output trajectories* $\boldsymbol{u}(t)$ *and* $\boldsymbol{y}(t)$ *over an interval* $0 \le t \le T$.

Definition Observability

4.3.1 Observability According to Kalman

To investigate the conditions for observability, one first considers the general solution for the output $\boldsymbol{y}(t)$ according to (3.56)

$$\boldsymbol{y}(t) = \boldsymbol{C}\Phi(t)\boldsymbol{x}_0 + \int_0^t \boldsymbol{C}\Phi(t-\tau)\boldsymbol{B}\boldsymbol{u}(\tau)\,\mathrm{d}\tau + \boldsymbol{D}\boldsymbol{u}(t) \tag{4.34}$$

with the transition matrix $\Phi(t)$ from (3.45). Since by assumption, in addition to the output $\boldsymbol{y}(t)$, the control input $\boldsymbol{u}(t)$ is known, the last two terms in (4.34) are known and thus do not affect the question of determining $\boldsymbol{x}_0$. To simplify the observability analysis, the input is set to $\boldsymbol{u}(t) \equiv 0$, so that (4.34) reduces to

$$\boldsymbol{y}(t) = \boldsymbol{C}\Phi(t)\boldsymbol{x}_0 \tag{4.35}$$

A (first) *necessary* condition for the observability of the system (4.33) can be derived by assuming that there exists a vector $\boldsymbol{a} \neq 0$ such that

$$\boldsymbol{C}\boldsymbol{A}^k\boldsymbol{a} = 0\,, \quad k = 1, 2, \ldots \tag{4.36}$$

is fulfilled. In this case one obtains

$$\boldsymbol{y}(t) = \boldsymbol{C}\Phi(t)\boldsymbol{a} = \left(\sum\nolimits_{k=0}^{\infty} \boldsymbol{C}\boldsymbol{A}^k \frac{t^k}{k!}\right)\boldsymbol{a} = 0 \tag{4.37}$$

However, this would imply that for the output $\boldsymbol{y}(t) = 0$ two different initial states correspond: $\boldsymbol{x}_0 = 0$ and $\boldsymbol{x}_0 = \boldsymbol{a} \neq 0$. Since this contradicts Definition 4.3, the *non-existence* of a vector $\boldsymbol{a} \neq 0$ satisfying (4.36) is a *necessary* condition for the observability of system (4.33).

This consideration can be further refined. Since according to Theorem 3.1 every matrix power $\boldsymbol{A}^n, \boldsymbol{A}^{n+1}, \ldots$ can be expressed as a linear combination of its lower powers $\boldsymbol{A}^0, \ldots, \boldsymbol{A}^{n-1}$, (4.36) reduces to

$$\boldsymbol{C}\boldsymbol{A}^k\boldsymbol{a} = 0\,, \quad k = 0, 1, \ldots, n-1 \quad \text{or} \quad \begin{bmatrix} \boldsymbol{C} \\ \boldsymbol{C}\boldsymbol{A} \\ \vdots \\ \boldsymbol{C}\boldsymbol{A}^{n-1} \end{bmatrix} \boldsymbol{a} = 0\,. \tag{4.38}$$

Condition for Observability according to Kalman

If the matrix in (4.38) has rank n, then there exists *no* vector $\boldsymbol{a} \neq 0$ that satisfies the equations in (4.36) or (4.38). This is at least a *necessary* condition for the observability of the system (4.33).

Observability Matrix

Theorem 4.6 (Observability According to Kalman)**.** *The system* (4.33) *is (completely) observable if and only if the* observability matrix

$$\boldsymbol{Q}_B = \begin{bmatrix} \boldsymbol{C} \\ \boldsymbol{C}\boldsymbol{A} \\ \vdots \\ \boldsymbol{C}\boldsymbol{A}^{n-1} \end{bmatrix} \quad \textit{or} \quad \boldsymbol{Q}_B = \begin{bmatrix} \boldsymbol{c}^\mathsf{T} \\ \boldsymbol{c}^\mathsf{T}\boldsymbol{A} \\ \vdots \\ \boldsymbol{c}^\mathsf{T}\boldsymbol{A}^{n-1} \end{bmatrix} \tag{4.39}$$

has rank n.

Exercise 4.4. *Calculate the observability matrix for the following system, and state whether the system is observable:*

$$\dot{\boldsymbol{x}} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & -3 & -2 \end{bmatrix} \boldsymbol{x} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \boldsymbol{x}.$$

To show that Theorem 4.6 is also *sufficient* (i.e. that the rank condition $\mathrm{Rang}(\boldsymbol{Q}_B) = n$ implies observability), the corresponding *Gramian Matrix* (observability gramian)

Gramian Matrix

$$\bar{\boldsymbol{G}} = \int_0^\mathsf{T} \left(\mathrm{e}^{\boldsymbol{A}\tau}\right)^\mathsf{T} \boldsymbol{C}^\mathsf{T}\boldsymbol{C}\,\mathrm{e}^{\boldsymbol{A}\tau}\mathrm{d}\tau \tag{4.40}$$

with $\Phi(t) = e^{\boldsymbol{A}t}$ is introduced. As in the case of controllability, one can show that $\bar{\boldsymbol{G}}$ is nonsingular if the observability matrix $\boldsymbol{Q}_B$ has rank n. Multiplying (4.35) from the left by $\bar{\boldsymbol{G}}^{-1}\left(\mathrm{e}^{\boldsymbol{A}t}\right)^\mathsf{T}\boldsymbol{C}^\mathsf{T}$ and integrating over the time interval $[0, T]$ yields

$$\int_0^T \bar{\boldsymbol{G}}^{-1}\left(\mathrm{e}^{\boldsymbol{A}t}\right)^\mathsf{T} \boldsymbol{C}^\mathsf{T}\boldsymbol{C}\,\mathrm{e}^{\boldsymbol{A}t}x_0\,\mathrm{d}t = \bar{\boldsymbol{G}}^{-1} \underbrace{\int_0^T \left(\mathrm{e}^{\boldsymbol{A}t}\right)^\mathsf{T} \boldsymbol{C}^\mathsf{T}\boldsymbol{C}\,\mathrm{e}^{\boldsymbol{A}t}\mathrm{d}t}_{\bar{\boldsymbol{G}}}\, x_0 = x_0 \tag{4.41}$$

thus providing an explicit procedure for computing $\boldsymbol{x}_0$.

Exercise 4.5. *Derive the rank condition for the observability matrix* (4.39) *by attempting to reconstruct the state $\boldsymbol{x}(t)$ of the system*

(4.33) *at time t from the knowledge of the outputs* $\boldsymbol{y}(t)$, $\boldsymbol{y}^{(i)}(t)$, $i = 1, \ldots, n-1$ *and the inputs* $\boldsymbol{u}(t)$, $\boldsymbol{u}^{(i)}(t)$, $i = 1, \ldots, n-1$.

Similarly to the issues encountered in the controllability analysis, the rank verification of the observability matrix $\boldsymbol{Q}_B$ for higher dimensional problems will generally become increasingly poorly conditioned. An alternative is therefore to verify the Gramian matrix (4.40):

Observability via the Gramian Matrix

Theorem 4.7 (Observability via the Gramian Matrix)**.** *The system* (4.33) *is (completely) observable if and only if the* Gramian Matrix (observability gramian)

$$\bar{\boldsymbol{G}} = \int_0^T \left(e^{\boldsymbol{A}\tau}\right)^{\mathrm{T}} \boldsymbol{C}^{\mathrm{T}} \boldsymbol{C} e^{\boldsymbol{A}\tau} \mathrm{d}\tau \tag{4.42}$$

is nonsingular. If the dynamics matrix $\boldsymbol{A}$ *is a Hurwitz matrix, then* $\bar{\boldsymbol{G}}_\infty = \lim_{T\to\infty} \bar{\boldsymbol{G}}$ *can be computed from the following Lyapunov equation:*

$$\boldsymbol{A}^{\mathrm{T}} \bar{\boldsymbol{G}}_\infty + \bar{\boldsymbol{G}}_\infty \boldsymbol{A} + \boldsymbol{C}^{\mathrm{T}} \boldsymbol{C} = 0\,. \tag{4.43}$$

If $\boldsymbol{A}$ is a *Hurwitz matrix*, then the existence of the integral (4.42) for the limit $T \to \infty$ is guaranteed. In Matlab the observability matrix $\boldsymbol{Q}_B$ and the Gramian matrix $\bar{\boldsymbol{G}}_\infty$ can be computed using the functions obsv and gram.

Question 4.4. *Formulate the property of observability in your own words.*

Example 4.4 (Hot Air Balloon [1])**.** *Figure 4.3 shows a hot air balloon, which can be approximately described by the differential equations*

$$\begin{aligned} \dot{h} &= v \\ \dot{v} &= -\frac{1}{\tau_1}(v - w) + k_1 \vartheta \\ \dot{\vartheta} &= -\frac{1}{\tau_2}\vartheta + k_2 u \end{aligned} \tag{4.44}$$

with the time constants $\tau_1, \tau_2 > 0$ *and the physical constants* $k_1, k_2 > 0$. *Here, h represents the altitude relative to a reference height, v is the vertical velocity, w is the vertical wind speed (thermal updraft), and* ϑ *is the temperature difference from the equilibrium temperature. The control input u represents the heat supplied by the burner.*
Since the wind speed w is an external disturbance, it is of interest whether both the temperature change ϑ *and a constant w can be estimated when only the altitude h is measured. For this purpose, the model* (4.44) *is augmented with the equation*

$$\dot{w} = 0 \tag{4.45}$$

and the observability matrix $\boldsymbol{Q}_B$ *is computed for the augmented system* (4.44), (4.45) *with the state* $\boldsymbol{x} = [h, v, \vartheta, w]^{\mathrm{T}}$ *and the measurement*

$y = \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} = [1, 0, 0, 0]\boldsymbol{x}$:

$$\boldsymbol{Q}_B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -\frac{1}{\tau_1} & k_1 & \frac{1}{\tau_1} \\ 0 & \frac{1}{\tau_1^2} & -k_1\frac{\tau_1+\tau_2}{\tau_1\tau_2} & -\frac{1}{\tau_1^2} \end{bmatrix} \quad \textit{with} \quad \det(\boldsymbol{Q}_B) = \frac{k_1}{\tau_1\tau_2}\,.$$

The observability matrix $\boldsymbol{Q}_B$ *is nonsingular for* $k_1 \neq 0$, *and thus the augmented system is completely observable.*

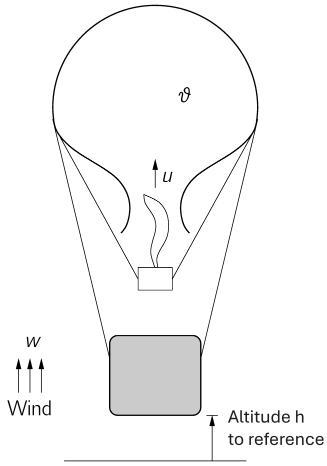

Figure 4.3: Hot air balloon from Exercise 4.4.

Exercise 4.6 (Cart-Pendulum System)**.** *Examine the observability of the linearized cart-pendulum model* (2.6) *for the outputs (see Figure 2.3)*

$$\Delta y_1 = \Delta x_w \quad \textit{or} \quad \Delta y_2 = \Delta\phi\,.$$

4.3.2 Observer Canonical Form (Single-Input Case)

By analogy to Section 4.2.2, in the following we consider a linear single-input system

$$\begin{aligned} \dot{\boldsymbol{x}} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}u\,, \quad \boldsymbol{x}(0) = \boldsymbol{x}_0 \\ y &= \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} + du \end{aligned} \tag{4.46}$$

with state $\boldsymbol{x} \in \mathbb{R}^n$, scalar input $u \in \mathbb{R}$, and scalar output $y \in \mathbb{R}$.
This section will show how a linear system (4.46) in state space can be transformed via a nonsingular state transformation

$$\boldsymbol{x} = \boldsymbol{V}\tilde{\boldsymbol{x}} \tag{4.47}$$

into the Observer Canonical Form

Observer Canonical Form

$$\dot{\tilde{x}} = \underbrace{\begin{bmatrix} 0 & \dots & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ \vdots & 1 & \ddots & \vdots & \vdots \\ 0 & 0 & \ddots & 0 & -a_{n-2} \\ 0 & 0 & \dots & 1 & -a_{n-1} \end{bmatrix}}_{\tilde{\boldsymbol{A}} = \boldsymbol{V}^{-1}\boldsymbol{A}\boldsymbol{V}} \tilde{x} + \underbrace{\begin{bmatrix} b_0 - a_0 b_n \\ b_1 - a_1 b_n \\ \vdots \\ b_{n-2} - a_{n-2} b_n \\ b_{n-1} - a_{n-1} b_n \end{bmatrix}}_{\tilde{\boldsymbol{b}} = \boldsymbol{V}^{-1}\boldsymbol{b}} u$$

$$y = \begin{bmatrix} 0 & \dots & 0 & 1 \end{bmatrix} \tilde{x} + b_n u \tag{4.48}$$

can be obtained. The observability matrix plays an important role here, as seen in the following theorem:

Transformation to Observer Canonical Form

Theorem 4.8 (Transformation to Observer Canonical Form)**.** *The linear time-invariant system (4.46) can be brought into observer canonical form (4.48) by the transformation (4.47) with*

$$\boldsymbol{V} = \begin{bmatrix} \boldsymbol{v} & \boldsymbol{A}\boldsymbol{v} & \dots & \boldsymbol{A}^{n-1}\boldsymbol{v} \end{bmatrix}, \quad \boldsymbol{v} = \boldsymbol{Q}_B^{-1} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, \quad \boldsymbol{Q}_B = \begin{bmatrix} \boldsymbol{c}^{\mathsf{T}} \\ \boldsymbol{c}^{\mathsf{T}}\boldsymbol{A} \\ \vdots \\ \boldsymbol{c}^{\mathsf{T}}\boldsymbol{A}^{n-1} \end{bmatrix} \tag{4.49}$$

provided that the observability matrix $\boldsymbol{Q}_B$ is nonsingular. Here, $\boldsymbol{v}$ represents the last column of the inverse observability matrix.

4.4 Duality Principle

In this learning unit you will learn to meaningfully combine the two previous sections on controllability and observability by using the duality principle.

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A scalar transfer function $G(s)$ remains unchanged when transposed, i.e.,

$$G(s) = \boldsymbol{c}^{\mathsf{T}} (s\boldsymbol{I} - \boldsymbol{A})^{-1} \boldsymbol{b} + d = \boldsymbol{b}^{\mathsf{T}} \left(s\boldsymbol{I} - \boldsymbol{A}^{\mathsf{T}}\right)^{-1} \boldsymbol{c} + d\,. \tag{4.50}$$

Motivated by this, one can generally define in the multi-input case ($m \geq 1$) the so-called *primal* and *dual* systems

primal and dual system

$$\begin{matrix} \text{primal} \\ \text{system} \end{matrix} \quad \begin{aligned} \dot{\boldsymbol{x}} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u} \\ \boldsymbol{y} &= \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u} \end{aligned} \iff \begin{aligned} \dot{\boldsymbol{x}}_D &= \boldsymbol{A}^{\mathsf{T}}\boldsymbol{x}_D + \boldsymbol{C}^{\mathsf{T}}\boldsymbol{u}_D \\ \boldsymbol{y}_D &= \boldsymbol{B}^{\mathsf{T}}\boldsymbol{x}_D + \boldsymbol{D}^{\mathsf{T}}\boldsymbol{u}_D \end{aligned} \quad \begin{matrix} \text{dual} \\ \text{system} \end{matrix} \tag{4.51}$$

For the transfer matrices it holds that $\boldsymbol{G}_D(s) = \boldsymbol{G}^{\mathsf{T}}(s)$, which can be easily shown by transposing $\boldsymbol{G}(s)$:

$$\begin{aligned} \boldsymbol{G}^{\mathsf{T}}(s) &= \left(\boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} + \boldsymbol{D}\right)^{\mathsf{T}} \\ &= \boldsymbol{B}^{\mathsf{T}}(s\boldsymbol{I} - \boldsymbol{A}^{\mathsf{T}})^{-1}\boldsymbol{C}^{\mathsf{T}} + \boldsymbol{D}^{\mathsf{T}} = \boldsymbol{G}_D(s)\,. \end{aligned} \tag{4.52}$$

Furthermore, it holds that the controllability (observability) of the primal system is equivalent to the observability (controllability) of the dual system. This is easily shown using the transposed controllability and observability matrices

$$\boldsymbol{Q}_S^\mathsf{T} = \boldsymbol{Q}_{B,D}\,, \quad \boldsymbol{Q}_B^\mathsf{T} = \boldsymbol{Q}_{S,D}$$

and is summarized in the following theorem.

Theorem 4.9 (Controllability/Observability of the Dual System)**.** *The dual system is controllable (observable) if and only if the primal system is observable (controllable).*

Literature

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4.5 Exercises

Training 4.1 (Asymptotical Stability). *Which of the following systems are asymptotically stable?*

$$(1)\quad \dot{x} = x\,, \quad x(0) = 0$$

$$(2)\quad \dot{\boldsymbol{x}} = \begin{bmatrix} 0 & 1 & 0 \\ -5 & -4 & 1 \\ 0 & 0 & -1 \end{bmatrix} \boldsymbol{x}\,, \quad \boldsymbol{x}(0) = \boldsymbol{x}_0$$

$$(3)\quad \dot{\boldsymbol{x}} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \boldsymbol{x}\,, \quad \boldsymbol{x}(0) = \boldsymbol{x}_0$$

$$(4)\quad \dot{\boldsymbol{x}} = \begin{bmatrix} -1 & 2 & -2 \\ -6 & 0 & 5 \\ -6 & 2 & 3 \end{bmatrix} \boldsymbol{x}\,, \quad \boldsymbol{x}(0) = \boldsymbol{x}_0$$

Give reasons for your statement in each case.

Training 4.2 (Controllability and Observability). *Investigate the complete controllability and complete observability of the system*

$$\begin{aligned} \boldsymbol{x} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}u\,, \qquad\qquad \boldsymbol{x}(0) = 0 \\ y &= \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} \end{aligned}$$

with the system matrices

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 \\ -5 & -4 & 0 \\ 0 & 0 & -2 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 0 \\ 1 \\ \alpha \end{bmatrix}, \quad \boldsymbol{c}^{\mathsf{T}} = \begin{bmatrix} 1 & 0 & \alpha \end{bmatrix}.$$

as a function of the real parameter α.

Training 4.3 (Transformation to Control and Observation Canocical Form). *Given is the dynamical system*

$$\begin{aligned} \dot{\boldsymbol{x}} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}u\,, \quad \boldsymbol{x}(0) = 0 \\ y &= \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} \end{aligned}$$

with the system matrices

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -2 & 3 \\ -3 & -2 & -1 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}, \quad \boldsymbol{c}^{\mathsf{T}} = \begin{bmatrix} 3 & 0 & 0 \end{bmatrix}.$$

1. *Check the complete controllability of the system.*
2. *Calculate the transformation matrix $\boldsymbol{T}$ that transfers the above system to the control normal form. Also specify the transformed system matrices.*
3. *Check the complete observability of the system.*
4. *Calculate the transformation matrix $\boldsymbol{V}$ that transforms the above system to observation normal form. Also specify the transformed system matrices.*

4.6 Solutions of Chapter 4

Solutions of Questions

Question 4.1:

By calculating and examining the eigenvalues of the dynamics matrix.

Question 4.2:

A system of dimension n is completely controllable if and only if the controllability matrix has rank n.

Question 4.3:

a) using the controllability matrix, b) using the Gramian matrix.

Question 4.4:

If a system is observable, then the initial state and thus the entire system state can be reconstructed or computed using the system outputs and, if necessary, the system inputs.

Solutions of Exercises

Exercise 4.1:

1. System: Equilibrium is not stable. 2. System: Equilibrium is not stable.

Exercise 4.2:

$\lambda_{1,2} = 0$, $\lambda_{3,4} = \pm\sqrt{\frac{6g(m+M)}{l(m+4M)}}$ $\Rightarrow$ the upper equilibrium is unstable

Exercise 4.3:

The controllability matrix is

$$\boldsymbol{Q}_S = \begin{bmatrix} \boldsymbol{b} & \boldsymbol{A}\boldsymbol{b} & \boldsymbol{A}^2\boldsymbol{b} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & -2 \\ 0 & -3 & 5 \end{bmatrix}$$

and has rank 3. Since the rank equals the dimension $n = 3$ of the system, the system is controllable.

Exercise 4.4:

The observability matrix is given by

$$\boldsymbol{Q}_B = \begin{bmatrix} \boldsymbol{c}^\mathsf{T} \\ \boldsymbol{c}^\mathsf{T}\boldsymbol{A} \\ \boldsymbol{c}^\mathsf{T}\boldsymbol{A}^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & -3 & -2 \\ -3 & 5 & 1 \end{bmatrix}$$

and has rank 3. Since the rank equals the dimension $n = 3$ of the system, the system is observable.

Exercise 4.5:

We start from a system of the form

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{B}\boldsymbol{u}(t) \tag{4.53}$$
$$\boldsymbol{y}(t) = \boldsymbol{C}\boldsymbol{x}(t) + \boldsymbol{D}\boldsymbol{u}(t). \tag{4.54}$$

Successive differentiation of (4.54) and substitution of (4.53) yields

$$\begin{aligned}
\dot{\boldsymbol{y}}(t) &= \boldsymbol{C}\dot{\boldsymbol{x}}(t) + \boldsymbol{D}\dot{\boldsymbol{u}}(t) = \boldsymbol{C}\boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{C}\boldsymbol{B}\boldsymbol{u}(t) + \boldsymbol{D}\dot{\boldsymbol{u}}(t)\\
\ddot{\boldsymbol{y}}(t) &= \boldsymbol{C}\ddot{\boldsymbol{x}}(t) + \boldsymbol{D}\ddot{\boldsymbol{u}}(t) = \boldsymbol{C}\boldsymbol{A}^2\boldsymbol{x}(t) + \boldsymbol{C}\boldsymbol{A}\boldsymbol{B}\boldsymbol{u}(t) + \boldsymbol{C}\boldsymbol{B}\dot{\boldsymbol{u}}(t) + \boldsymbol{D}\ddot{\boldsymbol{u}}(t)\\
&\vdots
\end{aligned}$$

The following system of equations is obtained to determine the vector $\boldsymbol{x}(t)$:

$$\begin{bmatrix} \boldsymbol{C} \\ \boldsymbol{C}\boldsymbol{A} \\ \boldsymbol{C}\boldsymbol{A}^2 \\ \vdots \\ \boldsymbol{C}\boldsymbol{A}^{n-1} \end{bmatrix} \boldsymbol{x}(t) = \begin{bmatrix} \boldsymbol{y} - \boldsymbol{D}\boldsymbol{u} \\ \dot{\boldsymbol{y}}(t) - \boldsymbol{C}\boldsymbol{B}\boldsymbol{u}(t) - \boldsymbol{D}\dot{\boldsymbol{u}}(t) \\ \ddot{\boldsymbol{y}}(t) - \boldsymbol{C}\boldsymbol{A}\boldsymbol{B}\boldsymbol{u}(t) - \boldsymbol{C}\boldsymbol{B}\dot{\boldsymbol{u}}(t) - \boldsymbol{D}\ddot{\boldsymbol{u}}(t) \\ \vdots \\ \vdots \end{bmatrix}$$

For this system to be solvable, the matrix on the left must have rank n.

Exercise 4.6 :

The system is observable with respect to Δy_1, but not observable with respect to Δy_2, which is directly evident from $\boldsymbol{Q}_{B,1}$ and $\boldsymbol{Q}_{B,2}$

$$\boldsymbol{Q}_{B,1} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{-3mg}{m+4M} & 0 & 0 & 0 \\ 0 & \frac{-3mg}{m+4M} & 0 & 0 \end{bmatrix}, \quad \boldsymbol{Q}_{B,2} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{6g(m+M)}{l(m+4M)} & 0 & 0 & 0 \\ 0 & \frac{6g(m+M)}{l(m+4M)} & 0 & 0 \end{bmatrix}.$$

Solutions of Trainings

Training 4.1:

(1) not asymptotically stable as $\text{Re}(\lambda) = 1 > 0$

(2) asymptotically stable as $\text{Re}(\lambda_i) < 0$ for all λ_i.

(3) not asymptotically stable as $\text{Re}(\lambda_{1,2}) = 0$

(4) not asymptotically stable as $\text{Re}(\lambda_1) = 5$

Training 4.2:

- **Controllability**: Checking the controllability matrix for full rank:

$$\text{Rang}(\underbrace{\begin{bmatrix} \boldsymbol{b} & \boldsymbol{A}\boldsymbol{b} & \boldsymbol{A}^2\boldsymbol{b} \end{bmatrix}}_{=Q_s}) \overset{!}{=} 3$$

  bzw.

$$\det(Q_s) \neq 0$$

- **Observability**: To analyse the observability, the observability tamrix Q_b must be considered instead of Q_s.

$$Q_b = \begin{bmatrix} \boldsymbol{c}^\mathsf{T} & \boldsymbol{c}^\mathsf{T}\boldsymbol{A} & \boldsymbol{c}^\mathsf{T}\boldsymbol{A}^2 \end{bmatrix}^\mathsf{T}$$

Result: For $\alpha \neq 0$ there is complete controllability/observability.

Training 4.3:

To 1.) The triangular shape of the controllability matrix

$$Q_s = \begin{bmatrix} 0 & 0 & 6 \\ 0 & 6 & -18 \\ 2 & -2 & -10 \end{bmatrix}$$

ensures full rank and therefore also controllability.

To 2.) Trasformation to control normal form $\tilde{\boldsymbol{x}} = \boldsymbol{T}\boldsymbol{x}$

$$\boldsymbol{T} = \frac{1}{6}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 3 \end{bmatrix}$$

$$\tilde{\boldsymbol{A}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -9 & -8 & -3 \end{bmatrix} \quad , \quad \tilde{\boldsymbol{b}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad , \quad \tilde{\boldsymbol{c}}^\mathsf{T} = \begin{bmatrix} 18 & 0 & 0 \end{bmatrix} \; .$$

To 3.) The triangular shape of the observability matrix

$$Q_b = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & -6 & 9 \end{bmatrix}$$

ensures full rank and therefore also observability.

To 4.) Trafo auf BNF $\boldsymbol{x} = \boldsymbol{V}\tilde{\boldsymbol{x}}$

$$\boldsymbol{V} = \frac{1}{9}\begin{bmatrix} 0 & 0 & 3 \\ 0 & 3 & -9 \\ 1 & -1 & -5 \end{bmatrix}$$

$$\tilde{\boldsymbol{A}} = \begin{bmatrix} 0 & 0 & -9 \\ 1 & 0 & -8 \\ 0 & 1 & -3 \end{bmatrix} \quad , \quad \tilde{\boldsymbol{b}} = \begin{bmatrix} 18 \\ 0 \\ 0 \end{bmatrix} \quad , \quad \tilde{\boldsymbol{c}}^\mathsf{T} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \; .$$